

AI/ML-BASED PERSONALIZED DIET PLAN GENERATOR FROM MEDICAL REPORTS

Abstract

Unhealthy dietary habits and lifestyle-related diseases such as diabetes, hypertension, and obesity are increasing rapidly. This project presents an AI/ML-based personalized diet plan generator that analyzes medical reports and doctor prescriptions to provide customized dietary recommendations. The system accepts inputs in the form of text, PDF, or scanned images. OCR and Natural Language Processing techniques are used to extract important medical information from unstructured reports. Machine learning models are applied to assess health risk levels based on clinical parameters. Based on the predicted risk and user dietary preferences, the system generates a personalized weekly diet plan along with foods to avoid, recommended foods, and dietary limits. The application is deployed using Streamlit, providing an end-to-end automated solution for personalized diet planning.

Introduction

In recent years, lifestyle-related health issues such as diabetes, hypertension, and obesity have become very common due to poor dietary habits and lack of physical activity. Doctors often provide dietary advice through handwritten or unstructured medical notes, which can be difficult for patients to understand and follow consistently. This creates a need for an automated system that can analyze medical information and provide clear, personalized diet recommendations.

With the advancement of Artificial Intelligence (AI) and Machine Learning (ML), it is possible to process medical data and generate meaningful insights. This project focuses on developing an AI/ML-based personalized diet plan generator that extracts relevant medical details from reports and predicts health risk levels. Based on these insights and user preferences, the system generates a structured and easy-to-follow diet plan, helping individuals maintain a healthier lifestyle.

Problem Statement

Medical reports and doctor prescriptions are usually written in unstructured formats, making it difficult for patients to clearly understand their health conditions and dietary requirements. Additionally, generic diet plans do not consider individual health parameters such as blood pressure, glucose levels, or body mass index. There is a need for an intelligent system that can automatically analyze medical reports, assess health risks, and generate personalized diet plans based on individual medical conditions and preferences.

Objectives

The main objectives of this project are:

- To analyze medical reports and doctor prescriptions provided in text, PDF, or image format
- To extract and structure important medical information using NLP and OCR techniques
- To predict health risk levels using machine learning models
- To generate personalized weekly diet plans based on medical conditions and user preferences
- To provide an easy-to-use web interface for diet plan visualization and report download.

System Architecture

The system follows a modular architecture where each component performs a specific task in the diet recommendation process. The architecture begins with user input, where medical data is provided in the form of text, PDF documents, or scanned images. OCR is applied to extract text from PDFs and images, while text inputs are processed directly.

The extracted text is cleaned and analyzed using Natural Language Processing techniques to identify key medical parameters such as blood pressure, glucose level, and BMI. These parameters are then passed to the machine learning model, which predicts the health risk level of the user. Based on the predicted risk and the extracted medical intent, a rule-based diet engine generates a personalized weekly diet plan. The final output is displayed through a Streamlit-based web interface, allowing users to download the diet plan in PDF and JSON formats.

Dataset Description

The dataset used in this project consists of medical and health-related attributes required for predicting health risk levels. It includes parameters such as age, glucose level, blood pressure, body mass index (BMI), insulin level, skin thickness, and other relevant clinical features. Multiple CSV files were collected and merged to form a single dataset for training and testing.

Before model training, the dataset was cleaned and preprocessed to handle missing values, remove inconsistencies, and normalize numerical features. The final dataset was then split into training and testing sets to evaluate the performance of different machine learning models. This dataset helps the system accurately predict health risks and support personalized diet plan generation.

Methodology

The proposed system follows a step-by-step methodology to generate personalized diet plans from medical reports. Initially, the user provides medical input in the form of text, PDF, or scanned images. For PDF and image inputs, OCR techniques are used to extract readable text from the documents.

The extracted text is then cleaned and processed using Natural Language Processing techniques to identify important medical parameters such as blood pressure, glucose level, and BMI. These extracted values are structured into a medical intent format, which represents the user's health condition in a machine-readable form.

Next, the structured medical data is passed to a machine learning model trained to predict the health risk level of the user. Based on the predicted risk and identified medical conditions, a rule-based diet engine generates a personalized weekly diet plan. The final diet plan is presented to the user through a web interface, with options to download the results in PDF and JSON formats.

Machine Learning Models Used

In this project, multiple machine learning models were explored to predict the health risk level of users based on clinical parameters. Initially, different classification models such as Random Forest, XGBoost, and LightGBM were implemented and evaluated. These models were chosen due to their ability to handle numerical medical data and capture complex relationships between health attributes.

After evaluation, the XGBoost model was selected as the final model due to its higher accuracy and better generalization performance. Hyperparameter tuning was performed to reduce overfitting and improve test accuracy. The trained model was saved and integrated into the application to provide real-time risk prediction based on user medical inputs.

AI / NLP Components

Artificial Intelligence techniques such as Optical Character Recognition (OCR) and Natural Language Processing (NLP) play a key role in this project. OCR is used to extract textual information from scanned medical prescriptions and PDF reports, enabling the system to handle unstructured inputs. NLP techniques are applied to clean the extracted text and identify relevant medical terms and numerical values.

The processed information is converted into a structured medical intent format, which includes identified conditions, measurements, and dietary advice. This structured representation allows seamless integration with the machine learning model and the diet recommendation engine, ensuring accurate and meaningful diet plan generation.

Backend & Deployment

The backend of the application is developed using Python, where all data processing, machine learning prediction, and diet plan generation logic are implemented. Streamlit is used as the frontend framework to build an interactive and user-friendly web interface. The application allows users to upload medical reports, select dietary preferences, and view personalized diet plans.

The complete project is version-controlled using GitHub, ensuring proper code management and collaboration. The application is deployed on Streamlit Cloud, enabling users to access the system through a web browser without requiring local installation. This deployment ensures scalability, ease of access, and real-time interaction with the AI/ML model.

Results & Evaluation

The performance of the machine learning models was evaluated using standard classification metrics such as accuracy, precision, recall, and F1-score. The dataset was divided into training and testing sets to ensure unbiased evaluation. Initial training results showed high training accuracy, indicating effective learning of patterns from the data.

However, signs of overfitting were observed, which were addressed through hyperparameter tuning and model optimization. After tuning, the XGBoost model achieved improved generalization performance with balanced training and testing accuracy. The final model demonstrated reliable risk prediction, which directly contributed to accurate and condition-specific diet plan generation. The deployed application was tested using multiple sample medical inputs, and the results confirmed correct risk classification and personalized diet recommendations.

Conclusion

This project successfully demonstrates the use of AI and machine learning techniques to generate personalized diet plans from medical reports. By combining OCR, NLP, and machine learning models, the system is able to analyze unstructured medical data and provide meaningful dietary recommendations. The integration of a user-friendly web interface and cloud deployment ensures easy accessibility and practical usability. Overall, the project offers an effective solution for personalized diet planning and highlights the potential of AI-driven healthcare applications.

